[0x3e]

$\cdots b_8 \mid b_7\, b_6\, b_5\, b_4\, b_3\, b_2\, b_1\, b_0$

$\oplus \quad b_8\, b_7\, b_6\, b_5\, b_4\, b_3\, b_2\, b_1$

1

$\Updownarrow$

$b_0 = 1$ OR $b_8 = 0$

& $b_1 = 0$ $\quad b_1 = 1$

block entropy

i.e. # of adjacent transition bits % 4

i.e. popcnt $(b_{i+1} \oplus b_i)$ # bits

i.e. $b_i \oplus b_{i+1} = 1$

$0\,0011\,1000$ (8)

$6\ 5\ \times 4\ 3\ \times 2\ 1$

$\wedge\ 0011$

11

binary → Gray code

mapping

```
mov     rdx, rax    rdx = rax_0
shr     rdx, 1      rdx = rax_0 >> 1
xor     rax, rdx
popcnt  rax, rax    rax = popcnt(rax_0 ⊕ (rax_0>>1))
and     rax, 0x3
```

means: $rax = rax_0 \oplus (rax_0 >> 1)$

popcnt: count # of bits set to 1

"entropy" of bits

like from David.M / @Volatility

i.e. $rax = popcnt\left(rax_0 \oplus (rax_0 >> 1)\right) \wedge 0x3$

{0,1,2,3}

# of adjacent transitions bits / 4

# of swaps between consecutive bits

4 3 2 1

0 0 1 0 0 1 1 0 → 4 bits

0 1 1 0 0 1 1 0

1 0 1 0 1 0 1 0

opposite of # of swaps

i.e. $f(0xa\ (1010)) = 0$ ← NO!

# of consecutive bit runs?

i.e. $f(0x3 = 0\cdots0011) = 1$ → ← NO!

$f(0x5 = 0\cdots0101) = 3$

[0x3e]

consecutive runs,

i.e. $f(0\cdots0110) = 2$, $f(0\cdots0001101011) = 3$?